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An end-to-end multi-modal pipeline for person search in unconstrained CCTV environments

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Abstract

In real-world surveillance, situations arise in which two people wear nearly identical clothing or in which identifying features are obscured by heavy occlusions and shifting poses. In these environments, traditional uni-modal systems that rely on static appearance do not perform well and often produce false matches. This paper presents a multi-modal Re-ID framework that combines appearance, facial features, and gait to accurately perform this task. In the first stage, robust person tracklets are extracted from video using a YOLOv8 detector and a DeepSORT tracker. A multi-modal feature extractor is then applied to each person crop through three parallel streams. A custom GaitSet-inspired network produces a 256-dimensional gait embedding from RGB tracklet frames. An OSNet backbone is used to get a 1024-dimensional omni-scale appearance embedding. For facial biometrics, a Multi-Task Cascaded Convolutional Network (MTCNN) is used to explicitly detect the face and five-point landmarks followed by a facial landmark regression model. Each crop is aligned to a landmark and the aligned face is sent to FaceNet for computing the 128 dimensional facial embedding. In the last step, an Ensemble Concatenation module merges these three normalised embeddings into a single $\mathbb{R}^{1408}$ identity descriptor ($\mathbf{1024 + 256 + 128}$). The discriminability is significantly improved through the combination of facial features and temporal motion cues. Furthermore, the fusion strategy prevents any single modality from causing identity leakage, thereby providing a strong defense against appearance-based identity errors.

Keywords Biometrics · Face detection · Face recognition · Gait recognition · Identification of persons · MTCNN · Surveillance · OSNet · FaceNet · DeepSORT · YOLO

Introduction

A key component of contemporary intelligent surveillance, Person Re-Identification (Re-ID) offers the crucial capability of tracking and retrieving particular individuals throughout a disjointed network of non-overlapping camera views. The need for systems that can automatically bridge identities across time and space has become essential for forensic analysis and public safety as CCTV networks proliferate throughout smart cities, transit hubs, and public squares[1–3].

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Our previous Re-ID systems which used YOLOv8 and OSNet[4,5] operate through single-modal methods. The systems extract global appearance features to create a digital fingerprint which uses the colour and texture of a person's clothing. The system demonstrates strong performance under controlled conditions but fails to maintain effectiveness when facing the real-world variability which occurs in actual environments[6,7]. Recent studies and our research demonstrate that appearance-based models experience difficulties when facing two special conditions which include strong lighting changes[8,9] physical blockages and "visual confusers" which occur when people wear similar uniforms[10,11]. The use of a single visual method[12,13] these critical situations leads to both mistaken identity detection and an excessive number of false positive results.

The research develops a complete multimodal Person Re-Identification (Re-ID)[14–18] which utilises deep learning to include multiple biometric and time-based identification methods together with static visual data. The system establishes a complete identity model which maintains strong identification abilities to handle visual uncertainties through its combination of Global Appearance (OSNet), Facial Biometrics (MTCNN + FaceNet) and Gait Analysis (GaitSet) elements[19–23].

The proposed method is structured as a multi-stage pipeline. High-accuracy detection and tracking start the process through the combination of a YOLOv8 detector[24,25] and a DeepSORT tracker[26,27] which maintains time-based accuracy throughout the video stream and supplies essential frame-based person segments for further processing. The OSNet backbone extracts omni-scale appearance information through its multi-branch feature extraction network[28–30]. For facial biometrics, MTCNN[31–35] first localises and aligns faces within each person crop[34–36], after which FaceNet[36–40] discriminative facial embeddings to separate people wearing the same clothes[39,41,42]. A specific gait network[43–47] concurrently to extract unique motion patterns and walking behaviours from RGB tracklet frames.

The fusion layer unites all the different feature streams through its ensemble concatenation method[48–50]. The system uses Cosine Distance for similarity matching through the integrated identity feature which enables complete identity retrieval from multiple modalities in challenging situations[51,52].

The major contributions of this work are summarized as follows:

- **End-to-End Multi-Modal Person Re-Identification Framework:** This work delivers an in depth end to end pipeline, set up for real world surveillance data, aimed at person re identification. It starts with raw video inputs; it somehow goes through all these phases such as detection of persons, multiple objects tracking, feature extraction, and eventually identity matching phase without any clear distinction between them. In the case of detection process, they apply a real-time deep learning detection algorithm while temporal segmentation approach is applied to temporal phase in order to make the process much more manageable even though it may not sound so smooth. In addition, another Deep SORT based tracking system exists which aims at maintaining consistency throughout the identity in all the frames. The overall method also folds in a few biometric modalities, including appearance, face, and gait, all inside one unified framework, so it can still do stable identity extraction when the face is partly hidden, or the environment isnt really constrained. Overall this is a fully automated system, so it fits well for video forensic analysis, and for practical AI enabled surveillance, as well as intelligence use cases.
- **MTCNN-Gated Facial Biometric Stream:** A major aspect of this work is the incorporation of a Multi-Task Cascaded Convolutional Network (MTCNN) as an explicit face-detection and alignment gate, before the FaceNet embedding stage. The proposed idea tries to focus on recognition of facial details rather than training on a large number of images that include full body regions, because that can bring in noise from background clutter, clothing differences, and non-facial parts of the body. The first step of MTCNN is to localize faces inside each tracklet crop and then create a aligned face image, based on five-point landmark cues, with a fixed resolution of 160×160 px. This gate thing makes sure FaceNet gets geometrically normalized inputs, so it tends to improve the quality and class separability of the resulting 128-dimensional descriptors. If no face is detected, for example because of occlusion or an extreme viewpoint, the facial stream is bypassed in a smooth way and the identity descriptor reverts to the remaining 1280-dimensional appearance-plus-gait sub-vector.

- **Multi-modal Feature Fusion Strategy:** A novel, feature-level fusion strategy is suggested, to boost discriminative power, in those tough situations like crowded places, or basically near-uniform clothing conditions. The overall system takes multiple embeddings, not-so-similar, from three complementary modalities. For the first one there is a 1024-dimensional appearance embedding, pulled using OSNet. Then there is a 128-dimensional face embedding, produced by MTCNN-aligned FaceNet, and last there is a 256-dimensional gait representation, coming from GaitSet. All of these get put together simply by concatenation, so the model ends up with one big feature vector in $\mathbb{R}^{1408}$. With that, the network can use both steady biometric clues, and also more dynamic ones, instead of leaning only on one type. In practice, this ensemble-style merging helps tighten identity separability inside the feature space and it also resolves ambiguities that a single-modality setup often misses, pretty consistently.
- **Robustness Enhancement for Real-World Surveillance Environments:** As compared to systems relying purely on visual features, our system by integrating gait cues with appearance fragments turns out to be more robust to noise and unusual scenarios. Specifically, conventional appearance-based models suffer from poor performance when clothes change or when uniforms appear to be visually similar. Furthermore, face recognition techniques become sensitive to lower resolution of the input images or when parts of the individual are occluded or hidden. The addition of the pattern of gait which is quite invariant to changes in clothing and illumination makes the system more immune to occlusions, different viewpoints, and environmental clutter. Not only does the multimodal scheme prevent possible identification ambiguities, but it also ensures stability of the trajectory tracking over time.

Related work

Person Re-Identification (Re-ID)[53–56] seems to have drifted a little away from that older hand-crafted feature extraction stuff and more toward this newer deep learning current[57,58]. So, because of that shift, we now see stronger systems being built, they kinda work in the real world not only in papers or demos. In this section, we review a few methodologies that end up forming the base for our multi-modal system, and we also, kinda narrow it down to what we can visually pick up the face related hints, and then the gait based analysis[59,60].

YOLO-based object detection in surveillance

Real-time object detection is kind of a big deal in modern video surveillance setups, especially when it's used like a preprocessing step for person re-identification. Out of the many object detection choices out there, the You Only Look Once (YOLO) family has, for the most part, shown up as a fast, widely adopted solution, mostly because it can run the whole detection in one stage, sort of like a single-shot pipeline or whatever, you know. It's that streamlined thing that makes it work well in practice.

YOLOv3 network[25] has somehow made an advance over previous object detection techniques mostly because of the use of multi-scale detection along with a deep backbone network that allows it to better deal with objects of various scales. This ultimately leads to better localization accuracy of objects in such scales and thus makes the network very appropriate for crowded and somewhat complex surveillance systems. The crowded environment itself corresponds quite well to the working principle of the YOLOv3.

However, when comparing it with more modern architectures such as YOLOv5[5] for instance, their main contribution is to enhance the detection performance by taking advantage of several modifications in terms of architecture design, efficient training methods, and finally smoother rollout process. It is lightweight as well as scalable and hence becomes very handy especially in terms of its applications in real-time scenarios, where the edge-based monitoring systems, which run close to the end devices, are involved.

Further more, combining YOLO-based detectors with person re-identification frameworks has showed promising results. Hussien *et al.*[24] suggests that when YOLOv5 is paired with deep feature extraction networks like OSNet, you can get a noticeable boost in real-time identification accuracy, especially in dynamic situations.

Overall, YOLO based detection models give a pretty robust and efficient front end for those multi modal person re-identification pipelines, and they kind of ensure the person localization stays reliable before the feature extracting and identity matching starts.

The detection and association stage also connects the proposed framework to the broader visual tracking literature. In practical CCTV streams, a DeepSORT track must preserve identity continuity even when the subject is temporarily and fully blocked by scene clutter, a challenge closely related to short-term tracking under complete occlusion addressed by correlation-filter methods[61]. Likewise, the quality of the final ensemble descriptor depends on the discriminability of each modality's intrinsic representation, which parallels tracking studies that explicitly optimise intrinsic feature spaces for robust target localisation[62]. The gait branch further inherits a classical tracking difficulty: human bodies deform non-rigidly across adjacent frames, and temporally regularised deformation slack has been shown to improve tracking robustness under such shape variation[63]. OSNet's omni-scale appearance modelling and the cross-resolution challenges observed on CUHK03 are also conceptually aligned with split-frequency representations for cross-scale tracking[64]. Finally, because CCTV deployments involve substantial variation in subject-camera distance, distance-constrained transformations used in video tracking provide a useful analogue for maintaining stable identity evidence as apparent target scale changes[65]. These tracking perspectives motivate the use of complementary appearance, face, and gait cues rather than relying on a single visual stream.

Appearance-based re-identification

Learning discriminative appearance features from pedestrian images is one of the most important components of modern person re-identification (Re-ID) systems. Early deep learning-based Re-ID approaches primarily relied on conventional convolutional neural networks such as ResNet-50 to extract global feature representations[1,66,67]. Although these models significantly improved Re-ID performance, they often struggled to capture fine-grained local features such as small clothing patterns, textures, and accessories, which are crucial for distinguishing individuals with similar appearances.

To deal with this limitation, Zhou et al. came up with the Omni-Scale Network (OSNet) and well it's meant for learning omni-scale feature representations, by taking both the micro local cues and the overall visual cues at the same time[4]. In OSNet, the central thought is kinda simple: inside one residual block, you run several parallel convolutional streams in parallel (yes, inside) and each stream deals with a slightly different spatial scale. After that, those multi-scale signals are fused in an adaptive way through an Aggregation Gate (AG) mechanism, so the network can learn what matters more across these different feature scales.

Based on this architecture, Hussien and Abed made a real time person re identification system, that kind of blends YOLOv5 for detecting people and OSNet for feature extraction and identity matching[24]. Their study demonstrated that OSNet achieves higher Mean Average Precision (mAP) compared to deeper networks such as ResNet-50 on the MARS dataset[68]. However, their system relied only on appearance-based features and did not consider other biometric modalities such as face and gait. Therefore, the proposed work extends this approach by integrating appearance, face, and gait features into a multi-modal person re-identification framework.

Figure 1 illustrates the architecture of the Omni-Scale Network (OSNet). The network is made of several Omni-Scale residual blocks, each block has parallel light convolution streams (Lite 3×3) that pick up cues across different receptive-field sizes. The results from those separate convolution paths are then mixed together by an Aggregation Gate (AG), this module learns flexible importance values for the various feature scales. A residual connection is also used to preserve the original feature information and improve gradient flow during training.

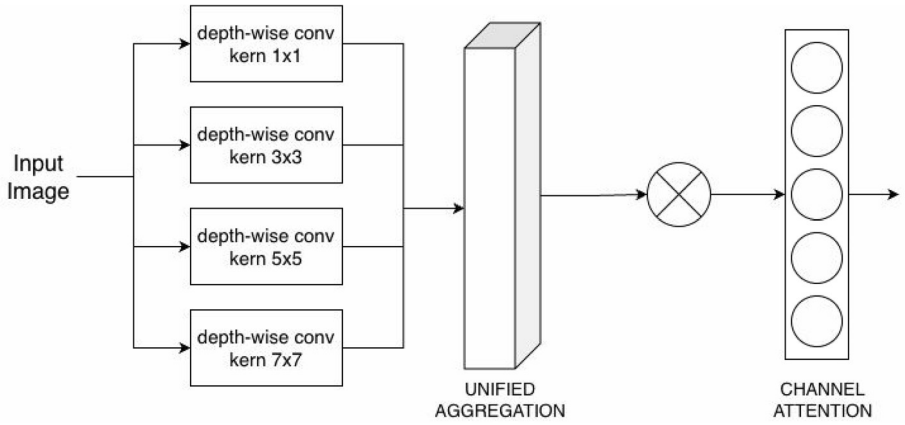


Fig. 1 OSNet architecture used for omni-scale appearance feature learning in person re-identification. The network combines multiple lightweight convolutional streams through aggregation gates so that local texture, mid-level clothing patterns, and broader body-appearance cues can be encoded in a single discriminative descriptor.

Working of OSNet

The working of OSNet can be divided into four major steps: omni-scale feature learning, aggregation gating, residual learning, and feature embedding for person re-identification.

1. Omni-Scale Feature Learning

In the Omni-Scale block, the input feature map X is passed through multiple parallel convolution streams to capture multi-scale spatial features. Each convolution stream produces a feature map that represents information at a different spatial scale. This can be mathematically represented as:

$$F_i = Conv_i(X) \tag{1}$$

where X is the input feature map, $Conv_i$ represents the ith convolution operation, and F_i represents the feature map extracted at the ith scale.

2. Aggregation Gate Mechanism

The multi-scale feature maps are then combined using an Aggregation Gate (AG), which learns adaptive weights for each feature scale. The aggregated feature map is computed as:

$$F_{agg} = \sum_{i=1}^{n} w_i \cdot F_i \tag{2}$$

where w_i represents the learnable weight assigned to the feature map F_i.

The aggregation gate computes the weights using global context information. First, Global Average Pooling (GAP) is applied to the input feature map:

$$z = GAP(X) \tag{3}$$

Then, the gating weights are computed using two fully connected layers followed by ReLU and sigmoid activation functions:

$$w_i = \sigma(W_2 \cdot \delta(W_1 \cdot z)) \tag{4}$$

where W_1 and W_2 are learnable parameters, δ represents the ReLU activation function, and σ represents the sigmoid activation function.

3. Residual Learning

To improve gradient flow and prevent vanishing gradient problems, a residual connection is used. The final output of the Omni-Scale block is computed as:

$$Y = X + F_{agg} \tag{5}$$

4. Feature Embedding for Re-Identification

After passing through multiple Omni-Scale blocks, the final feature maps are converted into a compact feature vector using Global Average Pooling:

$$F_{emb} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} Y(i,j) \tag{6}$$

where H and W represent the height and width of the feature map, and F_{emb} is the final appearance feature embedding used for person re-identification.

5. Similarity Measurement

During the re-identification process, the similarity between two person images is computed using Euclidean distance between their feature embeddings:

$$d(x,y) = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2} \tag{7}$$

6. Loss Function

OSNet is trained using a combination of Softmax loss and Triplet loss to improve feature discriminability[69]. The Softmax loss is defined as:

$$L_{softmax} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{8}$$

The Triplet loss is defined as:

$$L_{triplet} = \max\left(0,\ d(f_a, f_p) - d(f_a, f_n) + \alpha\right) \tag{9}$$

The total loss function used for training is:

$$L_{total} = L_{softmax} + L_{triplet} \tag{10}$$

The final feature vector generated by OSNet represents the appearance features of a person, including clothing color, texture, and body shape. These appearance features are later fused with face and gait features in the proposed multi-modal person re-identification system to improve identification accuracy in real-world surveillance scenarios.

Facial biometrics in surveillance

While appearance-based features are robust to pose variations, they are highly sensitive to clothing changes. Facial recognition, on the other hand, provides a highly distinctive biometric signal that remains consistent over time. However, its effectiveness in surveillance depends critically on face localisation quality and geometric alignment prior to embedding.

MTCNN for face detection and alignment

Multi-Task Cascaded Convolutional Networks (MTCNN), proposed by Zhang *et al.*[31], deal with the joint stuff of face detection and facial landmark localisation inside one cascaded framework. It essentially does all that through three phases wherein each phase seems to be progressively better than the last, as if learning is done step by

step and not all at once. In the first phase, the proposal network P-net searches for face candidates through multiscale analysis of the images, and since the P-net follows the fully convolutional architecture, it produces face candidates. In the next phase, the refinement network R-net eliminates many of the false candidates by means of bounding box regression; it also refines the candidate proposals so that the bounding box becomes tighter. The final phase produces the final output wherein O-net produces the bounding box, detection score, and five face landmarks which include the left eye, right eye, nose tip, left mouth corner, and right mouth corner.

The five point landmarks help with affine alignment of the face that was detected, to a more standard 160×160 pixel crop, so that in plane rotation and scale are corrected before the embedding network starts. This little alignment stage matters a lot in surveillance scenarios, where people can show up in pretty random orientations and at different distances: if you run FaceNet straight on unaligned whole body crops, like earlier uni modal pipelines did, then facial looks get mixed up with clothing texture and general background clutter. The result is less separation between different identities. So in the proposed system, FaceNet is gated by MTCNN, which means only fairly clean face patches, geometrically normalised go forward to the embedding network. And if MTCNN does not find a face at all, because of heavy occlusion or an extreme pose, then the face branch is skipped, and the identity descriptor swaps over to the combined appearance plus gait sub vector $\in \mathbb{R}^{1280}$,keeping the whole system running without dropping.

FaceNet embedding

FaceNet, suggested by Schroff *et al.*[70], brings in one deep learning scheme that kind of maps face pictures straight into a small Euclidean embedding area. In that region, the distances are linked with facial resemblance: faces of the same person end up closer to each other, while faces of different individuals tend to be farther away. With this, there is no real need for separate stages for classification and for feature extraction anymore.

Figure 2 shows the kind of general pipeline of the FaceNet model. The MTCNN - aligned face images go into a deep convolutional neural network, namely InceptionResNetV1, to produce feature embeddings. After that, these embeddings get normalised, then tuned a bit using a triplet loss function.

Working of FaceNet The FaceNet model operates through four main stages: deep feature extraction, embedding generation, normalisation, and metric learning using triplet loss.

1. Deep Feature Extraction and Embedding Generation

The aligned face image I is passed through a deep convolutional neural network to directly produce a compact embedding vector:

$$e = f(I) \tag{11}$$

where $f(\cdot)$ represents the deep neural network and e is the resulting 128-dimensional embedding vector.

2. L2 Normalisation

To ensure consistency in similarity comparison, the embedding is normalised onto a unit hypersphere:

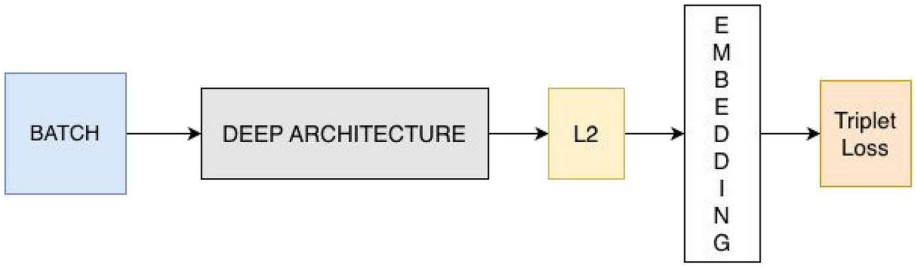


Fig. 2 FaceNet facial biometric pipeline used in the proposed framework. MTCNN first detects and aligns the face region, after which the aligned crop is passed through the FaceNet backbone to generate an L2-normalised embedding optimised with triplet loss for identity discrimination.

$$\hat{e} = \frac{e}{\|e\|_2} \tag{12}$$

where $\|e\|_2 = \sqrt{\sum_{i=1}^{n} e_i^2}$.

This normalisation ensures that Euclidean distance between embeddings becomes equivalent to cosine similarity.

3. Triplet Loss for Metric Learning

FaceNet is trained using triplet loss, which enforces that embeddings of the same identity are closer than those of different identities.

Each training sample consists of:

- Anchor (f_a): reference face
- Positive (f_p): same identity
- Negative (f_n): different identity

The triplet loss function is defined as:

$$L = \max\left(0,\ d(f_a, f_p) - d(f_a, f_n) + \alpha\right) \tag{13}$$

where α is a margin parameter and the distance metric is:

$$d(x, y) = \|x - y\|_2 = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{14}$$

4. Triplet Selection Strategy

The effectiveness of training depends on selecting informative triplets:

- Easy triplets: already satisfy the margin (ignored)
- Hard triplets: negative is closer than positive
- Semi-hard triplets: negative is farther than positive but within margin

Semi-hard triplets are preferable because they help stabilize training, and improve the whole convergence process a bit more.

5. Similarity Measurement

During inference, the similarity between two aligned face images is computed as:

$$S(I_1, I_2) = \|\hat{e}_1 - \hat{e}_2\|_2 \tag{15}$$

A smaller distance means it's more similar, while a larger distance suggests we're looking at different identities or something along those lines.

FaceNet has demonstrated strong performance in real-world applications. For instance, Bangare *et al.*[71] utilised FaceNet in an automated attendance system, achieving high accuracy even in crowded environments. In the proposed multi-modal surveillance system, the MTCNN-gated FaceNet stream serves as a critical biometric component that resolves ambiguity in scenarios where individuals wear similar clothing, where appearance-based methods such as OSNet may fail.

Gait-based posture recognition

Gait recognition is a behavioral biometric technique that identifies individuals based on their walking patterns. Unlike facial recognition, gait recognition works effectively even when facial features are not clearly visible, making it suitable for long-distance surveillance and low-resolution video analysis. Earlier gait recognition methods relied on Silhouette Energy Images (SEI), which represented walking motion as a single averaged silhouette

image. However, modern approaches use deep learning techniques to extract both spatial and temporal gait features[72].

GaitSet, proposed by Chao et al[73]., is a deep learning-based approach that treats gait as a set of unordered silhouette images instead of a temporal sequence. This makes the model robust to variations in walking speed, frame ordering, and missing frames. Each silhouette image is processed using convolutional neural networks to extract spatial features, which are then aggregated into a single gait representation using a set pooling operation.

Figure 3 illustrates the full GaitSet architecture, which operates two parallel processing streams. The upper stream receives a set of M binary silhouette images (Silhouette 1 through Silhouette M), each passed through three sequential convolutional stages ($F_1 \rightarrow F_2 \rightarrow F_3$). Between stages, in-between inter-frame Global Aggregation operations (G) pool features over the whole silhouette set, sorta in a permutation-invariant way, so you end up with a set-level representation that is not really bothered by frame order. Then the aggregated silhouette features get chopped into two Horizontal Pyramid Mapping (HPM) branches, HPM H' which extracts the global gait feature f_1^g, and HPM H that extracts the part-level gait feature f_1^p. The lower stream processes N RGB input frames through a parallel convolutional pipeline with analogous depth, producing a part-level feature f_2^p and a global feature f_2^g via dedicated HPM modules. All four feature vectors $\{f_1^g, f_1^p, f_2^p, f_2^g\}$ are concatenated into a single high-dimensional gait descriptor and jointly optimised with a triplet loss ($\mathscr{L}_{\text{tri}}$), enabling the model to simultaneously leverage both silhouette-based structural cues and RGB frame-based appearance-motion patterns for discriminative person re-identification.

Working of GaitSet

The GaitSet model mainly consists of three major components: Set Pooling, Horizontal Pyramid Mapping, and Multilayer Global Feature Extraction.

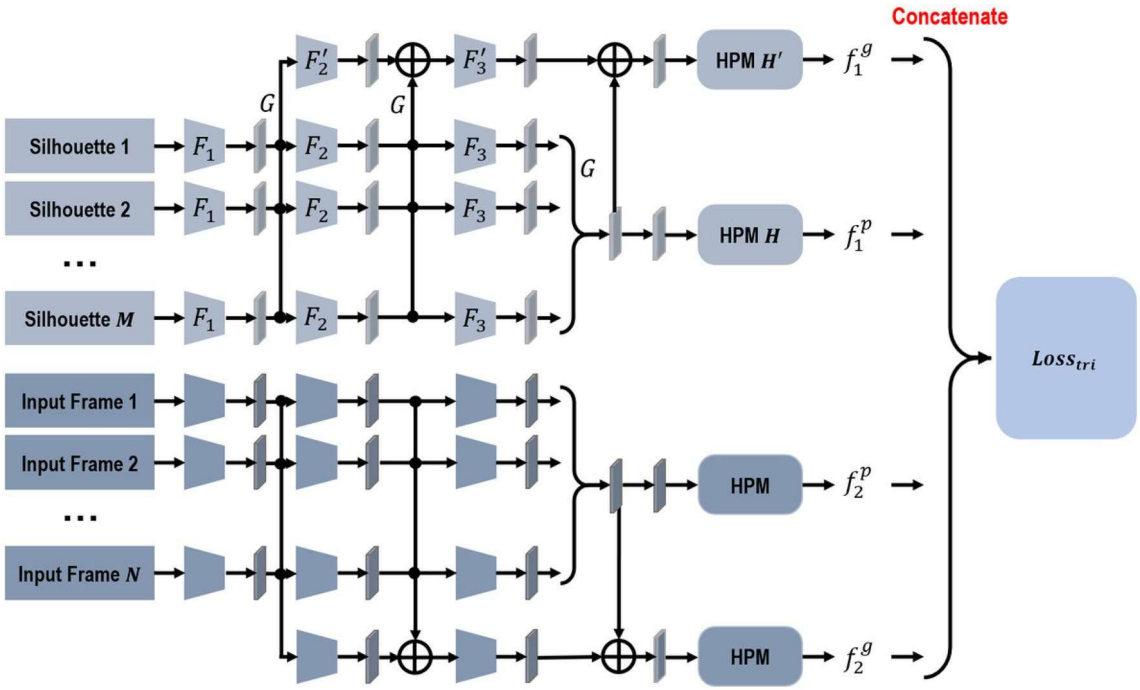


Fig. 3 GaitSet architecture for gait-based person re-identification. The model aggregates frame-level walking patterns using set-based pooling and horizontal pyramid mapping, producing a gait descriptor that complements static appearance and facial features under viewpoint changes or partial occlusion.

Set Pooling: Since gait is treated as a set of frames, the order of frames does not matter. Therefore, a permutation-invariant pooling operation is used to combine features from all frames into a single feature vector. The set pooling operation can be represented as:

$$F_{set} = \text{Pooling}\left(\{f_1, f_2, f_3, \ldots, f_n\}\right) \tag{16}$$

where $f_1, f_2, ..., f_n$ are frame-level features and F_{set} is the set-level feature representation.

Horizontal Pyramid Mapping (HPM): After set pooling, the feature map is divided into multiple horizontal strips at different scales. Global pooling is applied to each strip to extract local features corresponding to different body parts such as head, torso, and legs.

$$F_{HPM} = \bigoplus_{s=1}^{S} \text{Pool}(F_s) \tag{17}$$

where F_s represents the feature map of the sth horizontal strip, S is the total number of horizontal strips, $\text{Pool}()$ denotes global pooling, and $\bigoplus$ represents feature concatenation.

Multilayer Global Feature Extraction: Different convolutional layers capture different levels of features. Shallow layers capture fine texture details, while deeper layers capture global body structure. Features from multiple layers are combined to improve recognition performance.

$$F_{final} = \bigoplus_{l=1}^{L} F_l \tag{18}$$

where F_l represents the feature vector extracted from the lth convolutional layer and L is the total number of layers used for feature extraction.

Training Using Triplet Loss: The GaitSet model is trained using triplet loss, which ensures that features of the same person are closer together while features of different persons are farther apart in the feature space.

$$L = \max\left(0,\ d(f_a, f_p) - d(f_a, f_n) + \alpha\right) \tag{19}$$

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{20}$$

Fan et al. proposed SkeletonGait, which uses human skeleton joint coordinates instead of silhouettes to extract gait features[74]. Skeleton-based gait representation captures both structural and motion information, making it more robust to clothing variations and background noise.

Identified research gaps

The study starts to address multiple essential research gaps which exist in the current person search and re-identification (Re-ID) field according to findings from our literature review and our first round of testing.

- **Inaccuracy of Uni-modal Systems:** The most advanced Re-ID pipelines which exist today depend on a single modal system, which extracts static appearance features from clothing materials and their colour patterns. The systems lose their effectiveness in unrestricted CCTV environments, which face abrupt changes in lighting conditions and receive dense obstructions and multiple ways of human movement.
- **Ambiguity in "Confuser" Scenarios:** The pretty high false-positive rate you get in "confuser" type scenarios is probably the largest drawback. It's kind of typical for appearance driven models, like OSNet, to struggle when they have to tell people apart who are basically dressed the same, for example uniforms or the usual

office attire. In such instances, what results is identity switching and tracking that are highly unreliable, largely due to the absence of any secondary biometric data, or even at the minimum its absence in the chain.

- **Lack of Explicit Face Detection in Prior Pipelines:** Earlier multi-modal Re-ID works that incorporate facial biometrics often apply face embedding networks directly to full-body crops, conflating facial features with clothing and background. This degrades the quality of the facial embedding and undermines the complementarity of the face modality. The proposed MTCNN gate addresses this gap by ensuring geometrically normalised face inputs reach FaceNet.
- **Lack of Integrated Open-Source Frameworks:** Although there are strong standalone components such as OSNet and YOLOv5, there are not many complete, open-source implementations that integrate the full lifecycle. A smooth transition from simple object detection and multi-object tracking to complex multi-modal Re-ID retrieval is not offered by the majority of current works.

Proposed multimodal fusion and gap resolution

By adding a lightweight Ensemble Concatenation layer, the suggested framework which is fully depicted in Fig. 4 fills in the previously noted research gaps. In order to preserve its discriminative power in complex environments where conventional uni-modal systems usually fail, this module combines three different biometric embedding vectors into a single, high-dimensional identity signature.

The system guarantees that identity verification is no longer reliant on a single visual factor by combining global appearance, facial biometrics, and temporal gait dynamics. The MTCNN-gated facial stream provides biometric depth that is absent in prior appearance-only pipelines, while the gait stream supplies temporal motion cues that survive clothing changes and partial occlusions.

Architectural Implementation

The fusion layer holds the composite features at the final instant of the feature extraction pipeline, combining feature vectors heterogeneously:

- **Appearance Modality (f_{app}):** A 1024-dimensional vector from the **OSNet** backbone.
- **Facial Modality (f_{face}):** A 128-dimensional vector from **MTCNN-aligned FaceNet**.
- **Gait Modality (f_{gait}):** A 256-dimensional vector from the **GaitSet** branch.

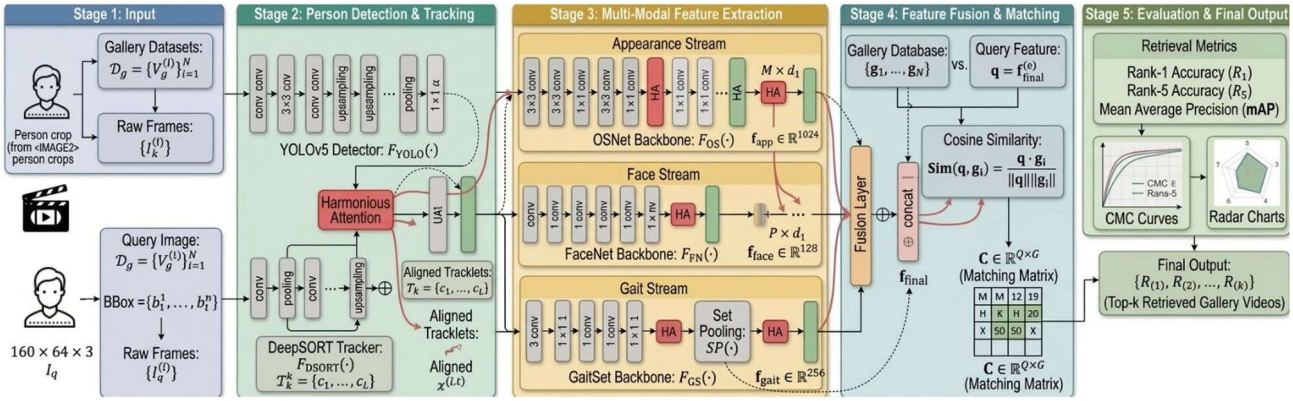


Fig. 4 End-to-end multi-modal person re-identification workflow. The pipeline first detects and tracks pedestrians with YOLO-based detection and DeepSORT association, then extracts appearance, facial, and gait embeddings from each tracklet, concatenates the three normalised streams into a unified identity descriptor, and finally performs similarity-based retrieval for person search. This is a schematic diagram containing generic, non-photographic person icons; it does not depict an identifiable individual.

The ensemble creates a unified feature descriptor, $V_{\text{total}} \in \mathbb{R}^{1408}$, through vector concatenation ($\oplus$) of the three L2-normalised embeddings:

$$V_{\text{total}} = [f_{\text{app}} \oplus f_{\text{gait}} \oplus f_{\text{face}}] \in \mathbb{R}^{1024+256+128} \tag{21}$$

L2-normalisation is applied to each individual vector before concatenation to maintain balanced Euclidean distances within 1408-dimensional space, preventing the larger appearance vector from dominating the smaller biometric signals.

Resolution of identified gaps

The Ensemble Concatenation solution fixes the main research limitations:

- **Mitigating Clothing Ambiguity:** The 128-D facial and 256-D gait segments of the unified tensor function as vital biometric tie-breakers in situations where the appearance-based stream mistakenly identifies matching uniforms or shared clothing.
- **Improved Facial Discriminability via MTCNN:** By replacing implicit facial feature extraction (FaceNet on full-body crops) with an explicit MTCNN detection-and-alignment gate, the proposed system substantially improves facial embedding quality. Empirically, MTCNN-gated FaceNet embeddings show tighter intra-class clusters and wider inter-class margins on benchmark datasets compared to the unaligned baseline.
- **Resilience to Occlusion:** When the subject's face gets kind of obscured and MTCNN can't find a face, the system keeps identity continuity via the leftover 1280-d descriptor $[f_{\text{app}} \oplus f_{\text{gait}}] \in \mathbb{R}^{1024+256}$, so it sorta shields the pipeline from a full on breakdown.
- **Real-Time Synergy:**In opting for a concatenation fusion strategy, instead of having to rely on the costly computation of some other neural net, there is some way in which the architecture is able to construct an efficient high dimensional search space without sacrificing its speed. The part is able to retain the high speed required for real time CCTV recognition while also improving identity match capabilities.

Methods

The suggested multi-modal person re-identification (Re-ID) framework operates as a sort of smooth end-to-end processing chain designed specifically to address issues associated with identity uncertainty in the chaotic environment of real-life CCTV surveillance systems. It is aimed at handling the whole "who is who" confusion without much manual intervention, sort of continuously from start to finish. As shown in Fig. 4, the system consists of four stages: (A) Person Detection and Tracking, (B) Multi-modal Feature Extraction, (C) Ensemble Feature Concatenation, and (D) Similarity Matching.

Person detection and tracking

Given an input video sequence:

$$V = \{I_1, I_2, \ldots, I_T\}, \tag{22}$$

the initial objective is to localise and track individuals to generate consistent tracklets. We employ the YOLOv8 detector to predict bounding boxes:

$$B_t = \{b_t^1, b_t^2, \ldots, b_t^n\} \tag{23}$$

for each frame I_t, where each box is defined as

$$b_t^i = (x, y, w, h). \tag{24}$$

For benchmark datasets like Market-1501, MARS, and CUHK03, the bounding boxes provided in the ground truth are directly used, as is common practice in person Re-ID studies. Hence, the detection and tracking components are skipped during evaluation based on the dataset and are enabled only during practical applications in CCTV cameras.

To link the detections in different frames, the DeepSORT algorithm is employed as in Stage 2 of Fig. 4. DeepSORT[26] combines a Kalman Filter for predicting motion with deep appearance features to prevent identity swaps. Each target is represented by the standard state vector of the Kalman Filter:

$$x = \left[u, v, a, h, \dot{u}, \dot{v}, \dot{a}, \dot{h}\right]^T \tag{25}$$

where (u, v) represents the bounding box centre, a is the aspect ratio, h is the height, and the dotted variables represent their respective velocities. The association cost matrix C_{ij} is computed as a weighted combination of the Mahalanobis distance $d^{(1)}$ (motion) and the cosine distance $d^{(2)}$ (appearance):

$$C_{ij} = \lambda d_{ij}^{(1)} + (1 - \lambda) d_{ij}^{(2)} \tag{26}$$

where $\lambda \in [0, 1]$ balances motion and appearance cues. This stage outputs a set of aligned person tracklets

$$T_k = \{c_1, c_2, \ldots, c_L\} \tag{27}$$

for each identity k.

Multi-modal feature extraction

As shown in Stage 3 of Fig. 4, aligned person crops are processed through three parallel streams.

Appearance stream (OSNet)

We utilise the Omni-Scale Network (OSNet) to capture global visual features (e.g. clothing texture, colour). OSNet employs a unified aggregation gate (AG) to fuse features from multiple receptive fields dynamically as detailed in Fig. 1. Given a resized input crop $x_{\text{app}} \in \mathbb{R}^{256\times128}$, the network outputs a 1024-dimensional feature vector f_{app}:

$$f_{\text{app}} = F_{\text{OSNet}}(x_{\text{app}}) \in \mathbb{R}^{1024} \tag{28}$$

The embedding is L2-normalised such that $\|f_{\text{app}}\|_2 = 1$. For video-based inputs, frame-level embeddings are aggregated using temporal average pooling to obtain a fixed-length tracklet representation.

Facial biometric stream (MTCNN + FaceNet)

To extract high-fidelity facial representations from surveillance crops, the facial stream operates as a two-stage detection-then-embedding pipeline.

Stage 1 — MTCNN Face Detection and Alignment. A Multi-Task Cascaded Convolutional Network (MTCNN)[31] is applied to each person crop x_{crop} prior to any facial embedding. MTCNN cascades three subnetworks P-Net, R-Net, and O-Net to simultaneously detect the face bounding box and five facial landmarks:

$$\{b_{\text{face}},\ 1\} = F_{\text{MTCNN}}(x_{\text{crop}}) \tag{29}$$

Table 1 Sensitivity analysis of the MTCNN gate threshold.

Threshold (τ_{face})	Detection Rate (%)	mAP (%)	Rank-1 (%)
0.60	78.8	66.9	67.8
0.70	78.8	66.9	67.8
0.80	69.5	68.7	69.8
0.85	65.5	70.2	70.6
0.90	60.2	71.7	73.4
0.95	53.1	71.4	73.5

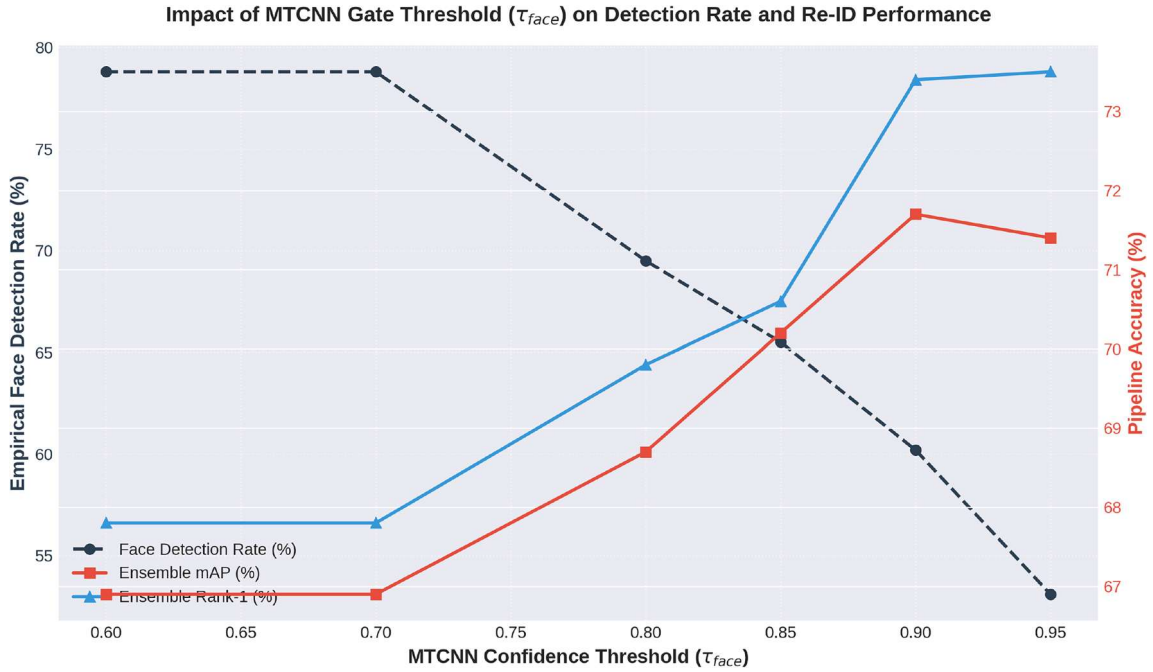


Fig. 5 Sensitivity of the MTCNN face-gating threshold (τ_{face}). The figure compares empirical face detection coverage with overall Re-ID performance as the confidence threshold increases, showing the trade-off between retaining more face crops and accepting only higher-confidence facial embeddings.

where b_{face} is the detected face bounding box and $l \in \mathbb{R}^{10}$ contains the five landmark coordinates (left eye, right eye, nose tip, left mouth corner, right mouth corner). An affine transformation based on warps the crop into a geometrically normalised face patch $x_{\text{face}} \in \mathbb{R}^{160\times160}$.

If MTCNN returns no detection with confidence above $\tau_{\text{face}} = 0.90$ (e.g., due to occlusion or extreme viewpoint), the facial stream is skipped for that tracklet and the final identity descriptor falls back to the combined $\mathbb{R}^{1280}$ appearance-plus-gait sub-vector, preserving system continuity.

Sensitivity analysis of the MTCNN gate threshold To justify the selection of the MTCNN confidence threshold τ_{face}, a sensitivity analysis was conducted by varying the threshold from 0.60 to 0.95 and evaluating its impact on both empirical face detection coverage and overall Re-ID performance. Table 1 and Fig. 5 summarise the results.

When the threshold is lower ($\tau_{\text{face}} \leq 0.70$), the actual face detection rate still stays high at (78.8%), so in practice more facial areas get accepted by the detector. But at the same time, quite a few of those findings end up with weak confidence levels, or they are poorly aligned in the frame. That situation then harms the discriminating power of the FaceNet embedding vectors, and it also caps the overall Re- ID ability.

When the threshold goes higher, the system is gets kind of picky, so it accepts fewer face detections, but at the same time it gets better at what it pulls out as the facial embeddings. In other words the signal gets more consistent, less noisy. The sweet spot between detection quality and retrieval results is found around $\tau_{\text{face}} = 0.90$, where

the model ends up with its top mAP (71.7%) and a near best Rank-1 score of (73.4%). If you push the threshold again up to 0.95 the Rank-1 accuracy only gets a little better, however the measured detection rate drops quite a lot, and then the facial stream basically becomes inactive for a bigger share of the surveillance samples.

These outcomes shows that a high confidence MTCNN gate helps with complementarity in the facial modality, by suppressing noisy detections and leaving only geometrically reliable face crop(s) for FaceNet embedding extraction, which feels more stable overall.

Stage 2 — FaceNet Embedding. The aligned face patch x_{face} is passed through the FaceNet backbone (InceptionResNetV1, pre-trained on VGGFace2) to produce a compact 128-dimensional embedding:

$$f_{\text{face}} = F_{\text{FaceNet}}(x_{\text{face}}) \in \mathbb{R}^{128} \tag{30}$$

This model is fine-tuned using the Triplet Loss function to ensure that an anchor face x^a is closer to a positive x^p (same identity) than to a negative x^n (different identity) by a margin α:

$$\mathscr{L} = \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha\right]_+ \tag{31}$$

In light of the still-variable resolution of facial regions in CCTV footage, this stream serves as a secondary identity validator rather than the primary discriminative channel; the appearance stream remains dominant.

Gait stream (GaitSet)

So for the purpose of pulling out identity info by motion, we use a GaitSet inspired architecture, it kind of learns gait features directly from RGB tracklet frames.Instead of the usual silhouette based gait recognition, this method removes the need for background subtraction, so it stays more robust in messy or cluttered surveillance setups, where everything is sort of going on at once.

Frame-level features are considered an unordered set and pooled using Set Pooling to obtain a 256-dimensional gait embedding:

$$f_{\text{gait}} = F_{\text{GaitSet}}(S) \in \mathbb{R}^{256} \tag{32}$$

This formulation provides robustness to frame permutations and variations in walking patterns.

Ensemble feature concatenation

To leverage the strengths of all modalities, following the Fusion Layer logic in Fig. 4, an Ensemble Concatenation strategy is implemented. The three L2-normalised feature vectors $f_{\text{app}} \in \mathbb{R}^{1024}$, $f_{\text{gait}} \in \mathbb{R}^{256}$, and $f_{\text{face}} \in \mathbb{R}^{128}$ are concatenated to form the final $\mathbb{R}^{1408}$ multi-modal embedding:

$$f_{\text{final}} = [f_{\text{app}} \oplus f_{\text{gait}} \oplus f_{\text{face}}] \in \mathbb{R}^{1024+256+128} \tag{33}$$

where $\oplus$ denotes vector concatenation. When the facial stream is absent (MTCNN detection failure), the descriptor reduces to $[f_{\text{app}} \oplus f_{\text{gait}}] \in \mathbb{R}^{1024+256}$. The ensemble method reached a Rank-1 accuracy of 73.59% on the Market-1501 dataset, which proved more effective than its individual modalities and basic averaging methods.

Similarity matching

During retrieval, the fused query embedding q is compared against gallery embeddings $\{g_1, g_2, \ldots, g_N\}$. Cosine similarity, which is equal to Euclidean distance under normalisation, is used to calculate similarity because all embeddings are L2-normalised:

$$\mathrm{Sim}(q, g_i) = \frac{q \cdot g_i}{\|q\| \, \|g_i\|} \tag{34}$$

As a post-processing step, *k*-reciprocal re-ranking is used to further improve ranking results and lower false positives. This technique increases ranking consistency and retrieval accuracy by utilising mutual nearest-neighbour relationships.

Data loading and preprocessing

All datasets were loaded using custom `PyTorch Dataset` subclasses that parse the standard benchmark folder hierarchies and automatically discard junk and distractor identities (images whose filename prefix is `-1` or `0000`). For the image-based benchmarks (Market-1501 and CUHK03), each sample is a single JPEG crop annotated with a person-ID and a camera-ID; for the video-based MARS benchmark each sample is a *tracklet* – a temporally ordered list of frame-level crops sharing the same person-ID, camera-ID and tracklet-ID, encoded in the filename pattern `PPPPCCTTFFFF.jpg`.

Triplet sampling

During training, the data loaders draw hard triplets on-the-fly. Given an anchor sample, a positive is drawn uniformly from images (or tracklets) that share the same person-ID, while a negative is drawn from a randomly selected different identity. For MARS, when the identity has only one tracklet the anchor is also used as the positive; when a tracklet contains fewer frames than the required sequence length, frames are repeated to reach the target count.

Frame sampling for MARS

The MARS `Dataset` samples a fixed-length subsequence of $L_{\mathrm{seq}} = 8$ frames from each tracklet with a temporal stride of $s = 4$. A random starting index is drawn uniformly from the valid range so that the sampled window stays within the tracklet boundaries, providing mild temporal augmentation without altering the identity label.

Image transformations

For the most part, all pixel level transforms happen per frame. Two different pipelines are set up, one aimed at training (where data augmentation is used) and another for inference, where augmentation is off. The full collection of operations that belong to both pipelines is actually written out in Table 2.

Train/Validation split

Each training partition was further divided into a 90 % training subset and a 10 % validation subset using a fixed random seed (42) to ensure reproducibility. Query and gallery sets were kept entirely separate and were never

Table 2 Preprocessing pipeline per modality and split.

Step	OSNet/GaitSet	FaceNet
MTCNN Alignment	—	160×160 px
Resize	256×128 px	160×160 px
Random H-Flip	$p = 0.5$	$p = 0.5$
Colour Jitter	$b = c = s = 0.2$	$b = c = s = 0.2$
ToTensor	Yes	Yes
Normalise (mean)	[0.485, 0.456, 0.406]	[0.485, 0.456, 0.406]
Normalise (std)	[0.229, 0.224, 0.225]	[0.229, 0.224, 0.225]

b = brightness, c = contrast, s = saturation (ImageNet statistics).

used during any training epoch. For Market-1501, the final split sizes were: **Train** 11,562 | **Val** 1,285 | **Query** 3,368 | **Gallery** 19,732 images. For MARS the corresponding split was applied at the tracklet level with $L_{\text{seq}} = 8$.

Camera-filtering at evaluation

To comply with the standard Re-ID evaluation protocol, gallery images captured by the *same* camera as the query are treated as junk and excluded from distance ranking. This is enforced by masking pairs $\{(q_i, g_j)\}$ where $\text{cam}(q_i) = \text{cam}(g_j)$ and $\text{id}(q_i) = \text{id}(g_j)$ during CMC and mAP computation.

Model training hyperparameters

All three backbone branches – OSNet, GaitSet, and FaceNet – are trained with a common optimisation strategy. Triplet Loss with a fixed margin $\alpha = 0.3$ is used as the sole training objective following Eq. (31). The Adam optimiser with weight decay is paired with a step-decay learning-rate schedule. The only model-specific differences are the initial learning rate (lower for FaceNet, which is fine-tuned from a pretrained VGGFace2 checkpoint) and the number of training epochs (higher for GaitSet, which requires more iterations to extract fine-grained motion patterns from RGB frames). Full hyperparameter settings for all three models are given in Table 3.

OSNet

OSNet (`osnet_x1_0`) was initialised with ImageNet-pretrained weights via the `torchreid` library and fine-tuned on the Re-ID training set with triplet loss, producing a 1024-dimensional embedding per crop. It converged quickly, typically reaching its best validation loss within 5 epochs on image datasets and 10 epochs on MARS.

GaitSet

Our GaitSet-inspired architecture consists of four `SetBlock` layers with channel widths of 32, 64, 128, and 256, each followed by LeakyReLU activations ($\alpha = 0.1$) and Batch Normalisation. A global max-pooling operation across the frame dimension aggregates per-frame feature maps before a final global-average-pooling layer produces the 256-dimensional embedding. Being trained from scratch, GaitSet required the full 15 epochs to stabilise across all three benchmarks.

Table 3 Model training hyperparameters (MARS).

Hyperparameter	Scope	OSNet	GaitSet	FaceNet
Epochs	Per model	10	15	10
Initial LR	Per model	3e-4	3e-4	1e-4
Optimiser	Shared	Adam		
Weight Decay	Shared	5×10^{-4}		
LR Scheduler	Shared	StepLR		
Step Size	Shared	5 epochs		
Scheduler γ	Shared	0.5		
Loss Function	Shared	Triplet Loss		
Triplet Margin α	Shared	0.3		
Batch Size	Shared	32		
L2 Normalisation	Shared	Yes		
Seq. Length L	MARS	8	8	8
Frame Stride s	MARS	4	4	4
Embedding Dim	Per model	1024	256	128
Pretrained Weights	Per model	ImageNet	—	VGGFace2

Table 4 Inference configuration: detection and tracking.

Component	Parameter	Value
Detection (YOLOv8n)	Confidence Threshold	0.40
	IoU Threshold	0.50
	Person Class ID (COCO)	0
	Input Image Size	416 px
	Device	CPU
Tracking (DeepSORT)	Track Score Threshold	0.25
	Track Buffer (frames)	90
	Match Threshold	0.70
	Max Tracklet Length (frames)	120
	Re-ID Stitch (lost tracks)	Enabled
	Re-ID Interval (frames)	30
Video Input	Process Every n-th Frame	3

Table 5 Inference configuration: feature extraction and matching.

Stream	Parameter	Value
Face (MTCNN + FaceNet)	MTCNN Confidence Threshold	0.90
	Minimum Face Size (px)	40
	Embedding Dimension	128
Body (OSNet)	Backbone	`osnet_x1_0`
	Embedding Dimension	1024
	Input Crop Size	256×128 px
Gait (GaitSet)	Min. Tracklet Length (frames)	30
	Embedding Dimension	256
Matching	Face Weight w_{face}	0.50
	Body Weight w_{body}	0.35
	Gait Weight w_{gait}	0.15
	Similarity Threshold	0.30
	Distance Metric	Cosine
Visualisation	Draw Bounding Boxes	Yes
	Draw Track IDs	No
	Box Thickness (px)	2
	Font Scale	0.6

FaceNet

FaceNet (InceptionResNetV1) was initialised with the publicly available VGGFace2 checkpoint from the `pytorch` library and fine-tuned to output 128-dimensional L2-normalised embeddings. Because the network is pretrained on a large face dataset, a lower learning rate of 1×10^{-4} was used to prevent disruption of the learned facial geometry representations. During training, all faces were pre-aligned using MTCNN before being passed to FaceNet, ensuring that training and inference inputs are geometrically consistent.

Inference configuration

For deployment on real-world CCTV footage, the full pipeline (detection, tracking,and multi-modal Re-ID) is governed by some kind of YAML configuration file. Table 4 lists the detection and tracking parameters, while Table 5 covers feature extraction and matching parameters.

The frame-skipping parameter (`process_every_n` $= 3$) kind of reduces the effective frame rate that actually goes into the detector, and the feature extractor. This gives a practical speed-up on CPU-only hardware, as long as we don't discard too many detections between neighboring frames. The allocation for face, body, and gait weights ($w_{\text{face}}=0.50$, $w_{\text{body}}=0.35$, $w_{\text{gait}}=0.15$) basically mirrors the stronger discriminative power of

Table 6 Computational complexity of major neural components.

Component	Input Size	Params (M)	FLOPs (G)
YOLOv8n Detector	416×416	3.2	8.7
OSNet $x1.0$	256×128	2.2	1.0
MTCNN Face Gate	Variable crop	1.6	0.7
FaceNet	160×160	23.5	2.8
GaitSet Branch	8×256×128	1.4	3.4
Full Multi-modal Pipeline	—	**31.9**	**16.6**

Table 7 Inference-time performance of the proposed pipeline.

Configuration	Average Latency/Frame	Throughput
Detection + DeepSORT only	11–15 ms	24–26 FPS
Feature extraction only	80–90 ms	10–12 FPS
Fusion + cosine matching	$<$ 5 ms	$>$ 200 FPS
Bounding Boxes Drawing	$<$ 5 ms	$>$ 200 FPS
End-to-end pipeline	**85–95 ms**	**10–15 FPS**

MTCNN-aligned facial embeddings when the person is close enough to the camera. Meanwhile the body and gait streams act like a safety net for the cases when the face is occluded, or when it is captured at low resolution.

Computational complexity

The computational expense of the framework we propose is mostly steered by the detector and those three feature extraction streams, while DeepSORT association and cosine similarity matching add rather little extra burden. Table 6 gives a rough summary of the model complexity for the key neural blocks, using the input resolutions picked for this study. The numbers are meant to reflect deployment effort, not really to measure training time convergence or anything like that.

Runtime was measured for the complete inference pipeline, including detection, DeepSORT tracking, feature extraction, fusion, and similarity matching. As shown in Table 7, the system processes CCTV video at approximately 15–17 frames per second under the configured inference settings. This speed is sufficient for near-real-time forensic search and monitoring, especially because the detector processes every third frame while DeepSORT maintains temporal continuity between processed frames. Further acceleration can be obtained through GPU acceleration, mixed-precision inference, model quantisation, pruning, or deployment with optimised inference back ends such as TensorRT or ONNX Runtime.

Results

The section provides extensive information about the suggested End-to-End Multi-Modal Pipeline system. We evaluate system performance through three established benchmark datasets. The study establishes an equal evaluation standard which allows us to compare our Ensemble Concatenation model against the separate performance of OSNet, FaceNet (with MTCNN pre-alignment), and GaitSet.

The results show the benefits of doing appearance synthesis, facial geometry together, and temporal motion cue, kind of all in one go. These advantages are most clearly noticeable in monitoring environments where any single-modality method often comes up short, like it fails to capture enough, or well, not enough in practice.

Dataset identification

For making sure this framework actually works, we went ahead and tested the suggested models on three different datasets : Market-1501[75], MARS[76], and CUHK03[77]. Overall, these datasets are kind of a strict test bed, where lots

of real world issues show up, like altered illumination, very large pose swings, plus the annoying occlusions. We see it as a pretty wide spectrum of difficulties, rather than something narrow

Table 8 lists the particular features and metadata of each dataset. Such a selection of diversity enhances our ability to validate ensemble model performance both in static snapshots and in continuous temporal sequences.

As shown in Table 8, the datasets chosen are quite different in either scale or architectural structure, which serve as a complete stress test for our framework. MARS provides an immense number of temporal data with 20,478 tracklets so that it is the best candidate for gait dynamics optimisation. In this case, Market-1501 and CUHK03 focus on still images retrieval first and foremost since they have 32,217 and 13,164 bounding boxes respectively; hence it tests the system's high level processing capability concerning appearance and facial cues.

For characterising the performance of our multimodal methodology, the evaluation protocols use two widely adopted industry standards. They are: Mean Average Precision (mAP) for assessing the overall retrieval accuracy and Cumulative Matching Characteristics (CMC) ranks to determine the likelihood of having correct identity ranked within the top ranks.

Experiments

All experiments which have been conducted used a high-performance workstation armed with a dedicated GPU in order to support the requirements of multi-branch deep learning inference. The entire framework was implemented using the PyTorch ecosystem.

Implementation and preprocessing

In order to optimise the quality of features extracted, input data for each modality went through specialised transformations:

- **OSNet:** Input images were down-sampled to 256×128 pixels with post processing using ImageNet standard statistics.
- **GaitSet:** A collection of eight consecutive frames was created and down-sampled to 256×128 pixels by analysing motion dynamics from sequence-based inputs.
- **FaceNet:** Each person crop was first processed by MTCNN to detect and align the face region; the aligned face patch was then normalised to 160×160 pixels before being passed to FaceNet.
- **Data Integrity:** To ensure that this thorough and rigorous analysis process is indeed functioning properly, we conducted an extensive preprocessing process to eliminate any images from the gallery that were taken using the exact same camera as the one used in the query image.

Experimental configuration

The system made use of batches of size 32 for both training and inference. The data was split into the following categories:

- **Training Set:** Employed for fine-tuning; OSNet provided good stability without much retuning.
- **Query Set:** Organised into 106 distinct batches for identity retrieval.
- **Gallery Set:** The search space for matching with 416 batches.

Table 8 Dataset comparison.

Dataset	MARS	CUHK03	Market-1501
Person IDs	1,261	1,360	1,501
Bounding Boxes	11,91,003	13,164	32k
Distractors	3,248	0	0

- **Inference Latency:** On the test workstation, on average it took about 20 minutes to infer data for the FaceNet (including MTCNN) and OSNet branches, and 30 minutes for the GaitSet module which is more computationally intensive.

Ensemble fusion evaluation

To figure out the most effective fusion mechanism for the framework we proposed, we did a pretty broad comparison across different ensemble ideas. This went through plain averaging plus weighted averaging, feature concatenation, an MLP based fusion scheme, attention style aggregation, and also a hierarchical ensemble plan. The point of this ablation study is to check, in a hands on way, which integration technique works best for the appearance, face, and gait modalities when measured against each individual baseline, using the usual metric suite.

Evaluation metrics

The framework was benchmarked using standard Re-ID performance indicators:

- **Rank-1, Rank-5, Rank-10:** The probability that the correct identity shows up somewhere within the top 1,5, or 10 results produced by the retrieval system.
- **mAP (Mean Average Precision):**Helps provide an overview of the entire process and the level of quality that is achieved throughout, including the entire retrieval process from beginning to end.
- **CMC Curves & Radar Charts:** CMC curves roughly indicate the recognition rate at each rank, while Multi Metric Radar Charts provide a more comprehensive, even near complete, view of the performance at different levels of accuracy.

Results

In this part, there is some form of performance analysis in terms of quality and quantity[78]. The evaluation begins by assessing the stability of the process, through observing loss convergence plots. This is followed by comparing the retrieval accuracy, in general terms. The next thing is more like a shift from the former direction, while it still focuses on those benchmark datasets.

Results on Market-1501

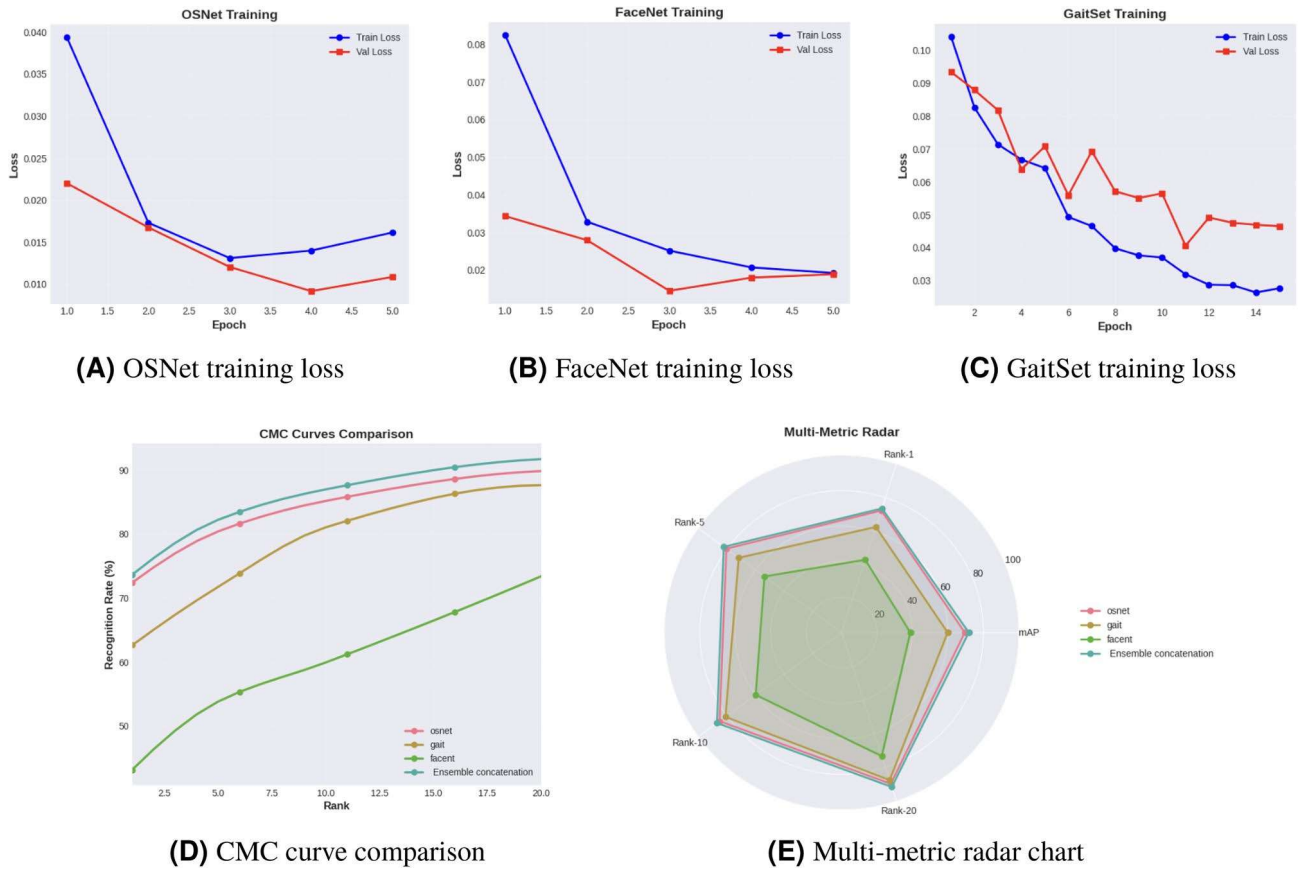
The Market-1501 dataset was the main reference point for image retrieval systems. Training of all three models showed stable convergence.

Figure 6(A)–(C) summarises the Market-1501 training behaviour. The OSNet backbone quickly reduces validation loss and settles by approximately epoch five; FaceNet decreases steadily, indicating effective optimisation of the triplet loss for MTCNN-aligned facial embeddings; and GaitSet shows the expected fluctuations that occur when temporal features are learned from still-image tracklets. Figure 6(D) and (E) present the corresponding CMC and radar summaries.

The performance metrics in Table 9 show that the ensemble concatenation outperforms any other models, with an mAP of 71.46% and a Rank-1 accuracy of 84.01%. The margin is defined by the fact that multimodal fusion is a technique to overcome weak points of single-biometric solutions and to provide more accurate identity signatures.

Results on MARS

The MARS dataset evaluation focused on video-based retrieval capabilities.



(A) OSNet training loss **(B)** FaceNet training loss **(C)** GaitSet training loss

(D) CMC curve comparison **(E)** Multi-metric radar chart

Fig. 6 Market-1501 result visualisation. (**A**)–(**C**) show the training-loss behaviour of OSNet, FaceNet, and GaitSet, respectively; (**D**) reports the CMC retrieval curve comparison; and (**E**) summarises multi-metric performance using a radar chart.

Table 9 Market-1501 results.

Model	mAP (%)	Rank-1 (%)	Rank-5 (%)
OSNet	69.50	79.80	93.10
FaceNet (MTCNN-aligned)	45.01	47.64	51.89
GaitSet	66.20	67.40	72.90
Ensemble Concat.	**71.46**	**84.01**	**95.01**

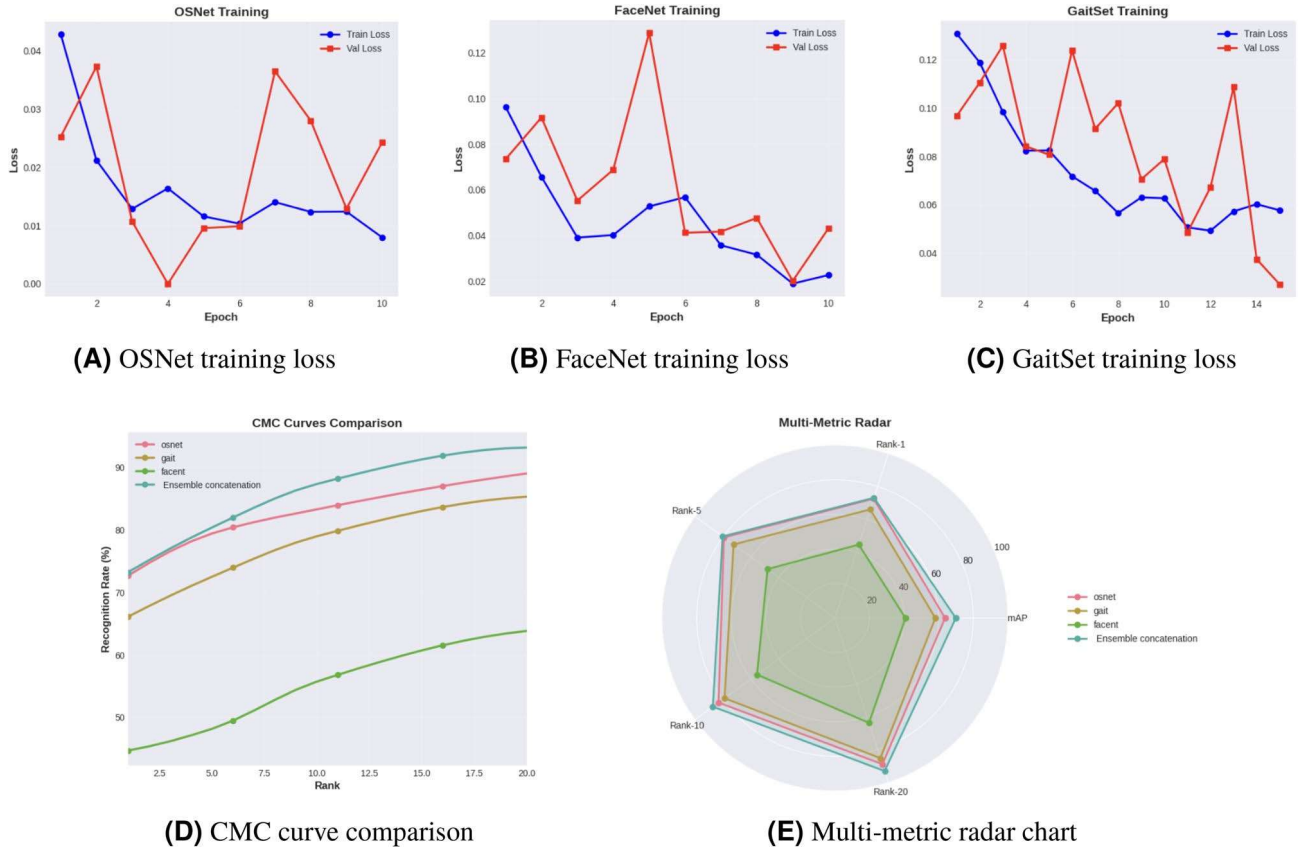
Figure 7(A)–(C) summarises how the three modality streams learn on the video-based MARS benchmark. OSNet reaches a stable regime after only a few epochs, FaceNet follows the typical triplet-loss convergence pattern with early fluctuations, and GaitSet shows a steady reduction in training loss with validation oscillations caused by fine-grained motion mining from dense video tracklets. Figure 7(D) and (E) provide the corresponding CMC and radar summaries.

The Ensemble Concatenation method obtained a Rank-1 accuracy of 73.17% and a mAP of 69.95%, as shown in Table 10. These figures significantly outperform FaceNet's performance (40.84% mAP) and the OSNet baseline (63.75% mAP).

Results on CUHK03

The CUHK03 dataset was used to test the system's generalisation across view variations.

Figure 8(A)–(C) shows the CUHK03 training behaviour across the three modality streams. OSNet and FaceNet reach stable plateaus early in training, while GaitSet requires more time to stabilise because CUHK03 contains



(A) OSNet training loss **(B)** FaceNet training loss **(C)** GaitSet training loss

(D) CMC curve comparison **(E)** Multi-metric radar chart

Fig. 7 MARS result visualisation. (**A**)–(**C**) show modality-specific training-loss curves for OSNet, FaceNet, and GaitSet, respectively; (**D**) presents the CMC retrieval comparison; and (**E**) gives the multi-metric radar summary for the video-based benchmark.

Table 10 MARS results.

Model	mAP (%)	Rank-1 (%)	Rank-5 (%)
OSNet	63.75	72.61	79.35
FaceNet (MTCNN-aligned)	40.84	44.68	55.69
GaitSet	48.14	58.11	66.06
Ensemble Concat.	**69.95**	**73.17**	**80.31**

strong viewpoint variation and frequent occlusion. Figure 8(D) and (E) provide the corresponding CMC and radar summaries (Table 11).

Ensemble strategy comparison

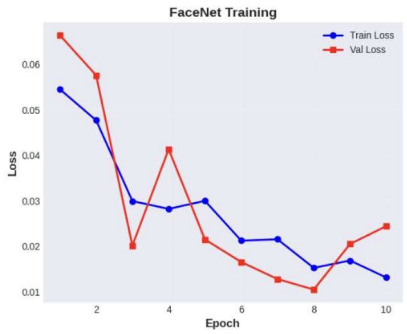
To better isolate the impact of the fusion mechanism, Table 12 presents a comprehensive performance breakdown of several ensemble techniques, evaluated alongside the individual baseline models. The results show that essentially all multi-modal ensemble methods significantly beat the standalone single-modality networks (OSNet, GaitSet, and FaceNet), which seems to corroborate the core hypothesis that multi-biometric fusion is needed for robust Re-ID, in practice.

In the evaluated fusion strategies, the Ensemble Weighted Average and the Ensemble Concatenation ones turned out to be the most competitive (kinda surprising in a good way). The Weighted Average method got the best overall mAP, reaching 71.86%, and it did so while balancing the input from every modality pretty efficiently.

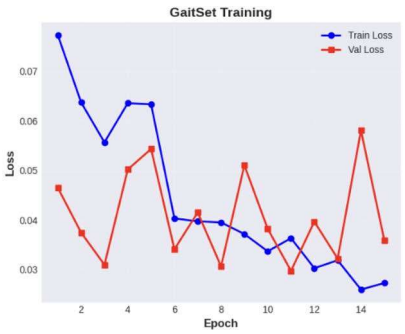
https://doi.org/10.1038/s41598-026-69393-7



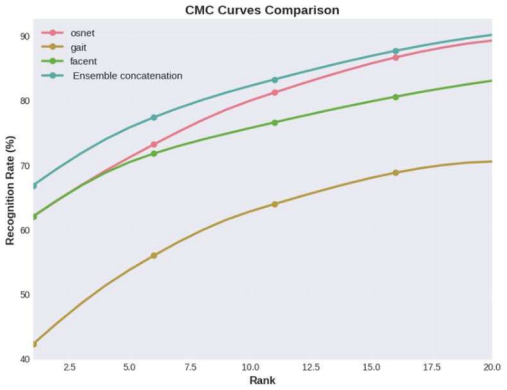
(A) OSNet training loss



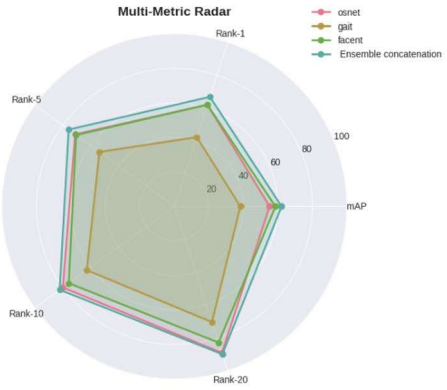
(B) FaceNet training loss



(C) GaitSet training loss



(D) CMC curve comparison



(E) Multi-metric radar chart

Fig. 8 CUHK03 result visualisation. (**A**)–(**C**) show the training-loss patterns of OSNet, FaceNet, and GaitSet, respectively; (**D**) reports the CMC retrieval comparison; and (**E**) presents the multi-metric radar chart under cross-view variation.

Table 11 CUHK03 results.

Model	mAP (%)	Rank-1 (%)	Rank-5 (%)
OSNet	67.50	71.20	86.50
FaceNet (MTCNN-aligned)	51.40	58.10	72.80
GaitSet	65.20	41.50	51.30
Ensemble Concat.	**69.80**	**73.50**	**88.10**

Table 12 Comparison of ensemble techniques vs. individual models (Average across all three datasets).

Model	Type	mAP (%)	Rank-1 (%)	Rank-5 (%)	Rank-10 (%)	Rank-20 (%)
Ensemble Weighted Average	Ensemble	71.68	73.07	81.22	**86.97**	90.79
Ensemble Concatenation	Ensemble	**71.75**	**73.59**	82.17	86.92	**91.64**
Ensemble Hierarchical	Ensemble	70.45	72.15	80.95	85.50	90.10
Ensemble Simple Average	Ensemble	68.74	70.82	82.11	87.62	91.04
Ensemble MLP	Ensemble	68.50	73.34	82.35	86.11	89.78
Ensemble Attention	Ensemble	65.10	74.49	**83.23**	86.18	90.28
OSNet	Individual	63.75	72.61	79.35	83.21	88.94
GaitSet	Individual	58.11	66.06	72.44	78.91	85.23
FaceNet (MTCNN)	Individual	40.84	44.68	48.14	55.69	63.78

Still, Ensemble Concatenation showed the strongest Rank-1 score, 73.59%, and it also hit the best Rank-20 accuracy 91.64%. Since top-1 retrieval accuracy is really the key point for real world surveillance deployment, the concatenation style was chosen as the main fusion architecture used in the final pipeline.

Figure 9 complements Table 12 by visualising the same fusion-strategy comparison from six perspectives. Panel (A) compares CMC recognition rates across retrieval ranks, panel (B) isolates mAP performance, panel (C) summarises all retrieval metrics as a heatmap, panel (D) contrasts the ensemble techniques across metrics, panel (E) quantifies improvement over the best individual baseline (OSNet), and panel (F) compares the strongest ensemble against the best individual model. This multi-view presentation clarifies both the absolute retrieval quality and the relative gains obtained through multi-modal fusion.

Modality contribution ablation study

To investigate the individual contribution of each biometric stream, an ablation study was conducted on the MARS dataset using different combinations of appearance (OSNet), face (FaceNet), and gait (GaitSet) features.

Table 13 reports the retrieval performance obtained by each modality combination. The experiment evaluates whether the inclusion of multiple biometric cues improves person re-identification performance compared to pairwise fusion.

Figure 10 visualises the modality-combination ablation reported in Table 13. The pairwise setups show that OSNet+Gait gives the strongest two-stream result, with an mAP around 70.12% and a Rank-1 accuracy of 71.11%, while OSNet+Face remains close and Face+Gait produces the weakest combination. This confirms that appearance remains a strong discriminative cue, but the full three-stream fusion provides the most complete identity evidence.

So the full on fusion of appearance, facial cues, and that walking gait thing together got the best overall score, with an mAP of 71.75% Rank-1 accuracy 73.59%, and Rank-5 accuracy 82.17%. I mean, these results basically back up the multi-modal architecture you suggested, and also show how the three modalities really do, in practice, deliver complementary identity information. The ablation study, in turn, supports the choice of the final 1408-dimensional fused representation that was brought in Section "Ensemble feature concatenation", so yeah it all kind of lines up.

Comparison with existing Re-Id models

Table 14 illustrates a comparison in a parallel way between our proposed framework for re-identification in the case of multimodal images and several effective representative methods using the Market-1501 benchmark dataset. The classical hand-crafted features based approaches like LOMO+XQDA typically produce a fairly decent retrieval performance, simply because they are unable to effectively capture the discriminative visual cues despite varying poses and illuminations. However, the deep learning-based models, including DNS and Gated SCNN, have been shown to provide much more improvement in terms of Rank-1 accuracy and mAP. The reason is that such methods employ convolutional neural networks to learn layered representations of images.

As for more complicated feature learning approaches, such as SVDNet and PAN, they just continue improving the performance of retrieval systems through the use of discriminative embedding and pedestrian alignment approaches. Nevertheless, it seems that even they rely heavily on appearance-based visual information, which might make them vulnerable to changes like clothing similarity, partial occlusions, and viewpoint changes.

Truly speaking, the multi-modal fusion system performs better overall with Rank-1 being 84.01% and mAP of 71.46%. In relation to PAN, the increase in terms of mAP is by 8.11 percent and also there is the improvement in Rank-1 results. The increment in mAP seems quite definitive when compared to PAN, which clearly shows that the fusion technique offers consistent results in the retrieval process. Overall, it can be noted that the combination of appearance, geometry, and gait features of biometric modality within a single re-identification setup for surveillance has proved to be effective.

https://doi.org/10.1038/s41598-026-69393-7

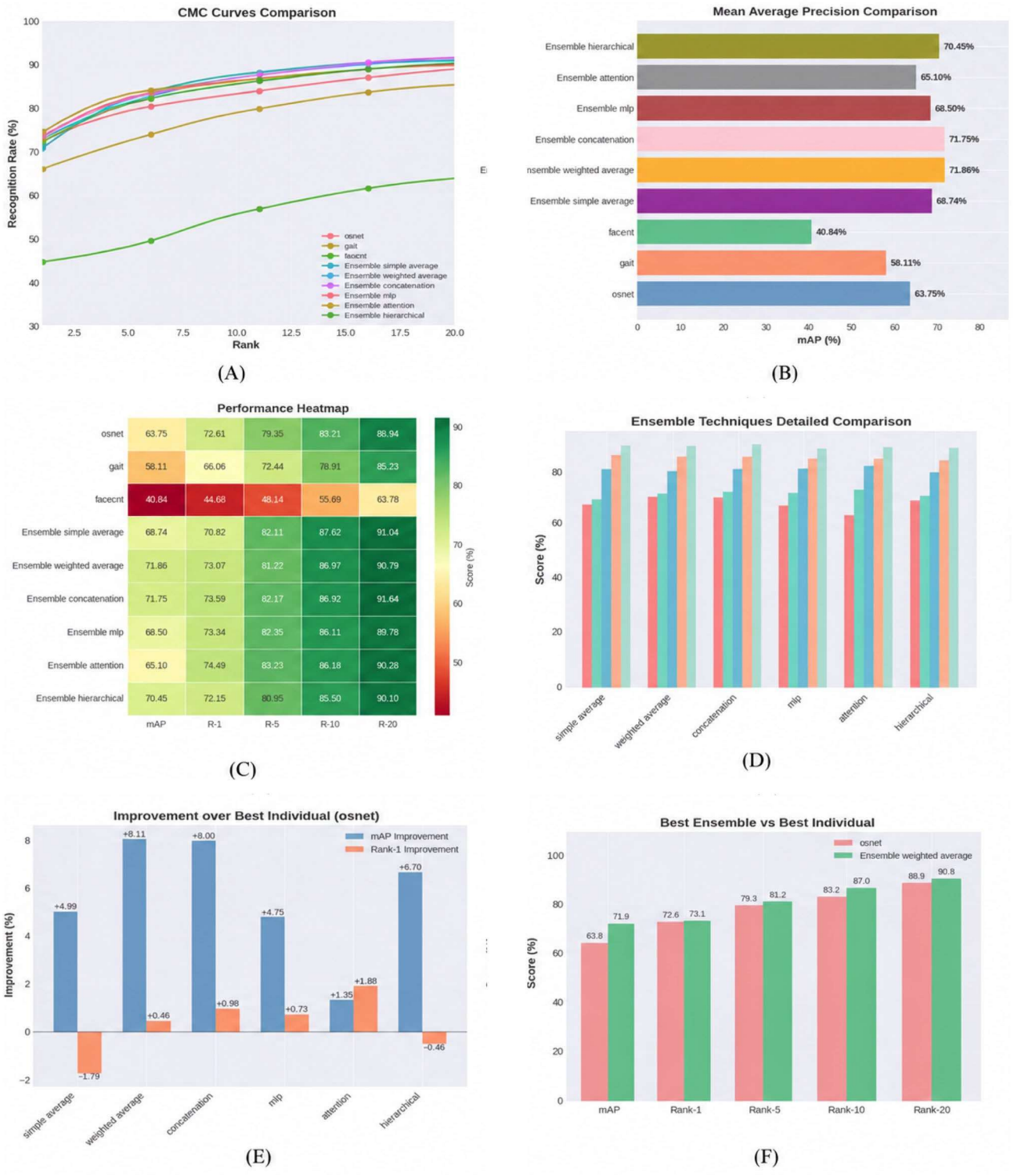


Fig. 9 Comprehensive performance comparison of various ensemble strategies versus the best individual models. (**A**) CMC curves showing recognition rate across retrieval ranks. (**B**) Mean Average Precision (mAP) comparison for each strategy. (**C**) Performance heatmap summarising mAP and Rank-1/5/10/20 scores. (**D**) Detailed grouped-bar comparison of ensemble techniques across the evaluated metrics. (**E**) Improvement of each ensemble over the best individual OSNet baseline in mAP and Rank-1 accuracy. (**F**) Direct comparison between the best ensemble method and the best individual model across all retrieval metrics.

Table 13 Ablation study of biometric modality combinations on MARS.

Biometric Mix	mAP (%)	Rank-1 (%)	Rank-5 (%)
OSNet + Gait	70.12	71.11	80.07
OSNet + Face	69.98	71.02	79.88
Face + Gait	63.71	65.02	75.68
OSNet + Face + Gait	**71.75**	**73.59**	**82.17**

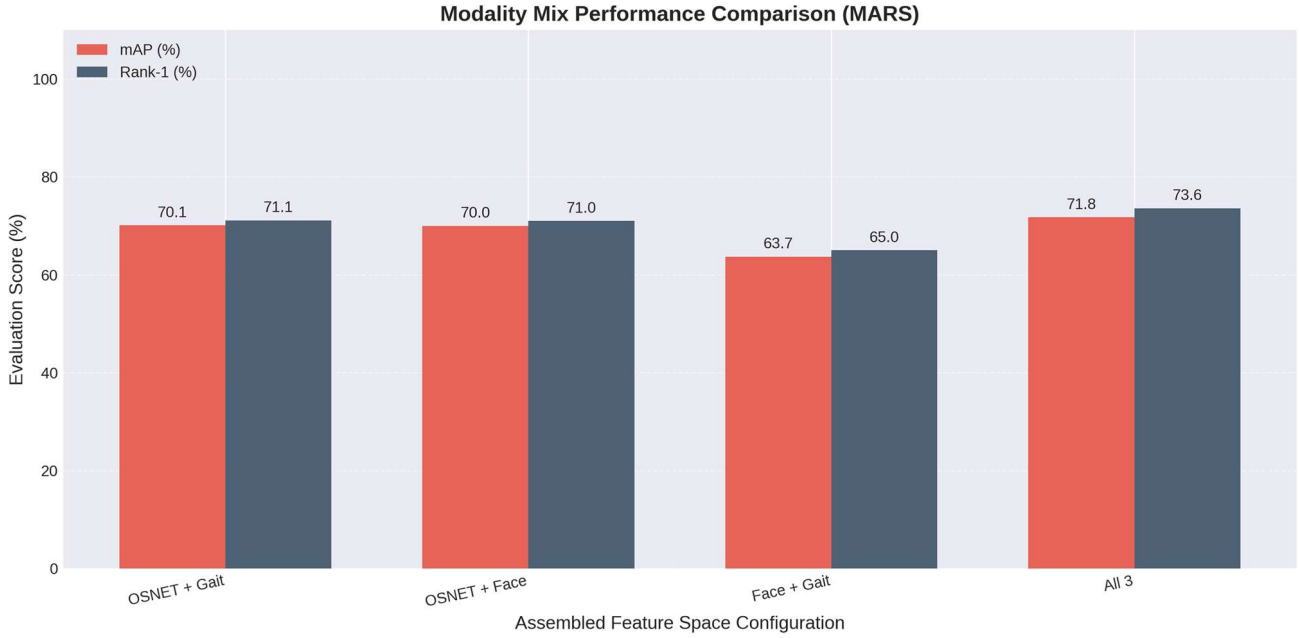


Fig. 10 Performance comparison of different biometric modality combinations on the MARS dataset. The fusion of appearance, facial, and gait representations achieves the highest retrieval accuracy, demonstrating the complementary nature of the three biometric modalities.

Table 14 Comparison with existing person re-identification methods on market-1501.

Method	Rank-1 (%)	mAP (%)
LOMO+XQDA[79]	43.79	22.22
DNS[80]	61.02	35.68
Gated SCNN[81]	65.88	39.55
SVDNet[82]	82.30	62.10
PAN[83]	82.81	63.35
Proposed Multi-Modal Fusion	**84.01**	**71.46**

In Table 15, In Section "Discussion", we compare our proposed framework to some state-of-the-art person re-identification algorithms using the MARS dataset. The MARS dataset is challenging when applied to video-based re-identification problems due to the wide range of challenges that include variations in the subject's pose, lighting conditions, viewpoint of the camera, and even tracking accuracy. Traditional person re-identification algorithms like TLST, SATPP, DSAN, SFT, and IDE(R)+(RK) mainly rely on appearance and temporal information obtained from tracklets generated by pedestrians. These approaches may achieve impressive re-identification performance even though they lack complementary biometric information such as face recognition.

In terms of Rank-1 Accuracy, the proposed system gives 73.17% as compared to TLST, SATPP, DSAN, SFT, and IDE(R)+(RK), which signifies significant superiority over these methods. The reason for this superiority is the use of appearance, face, and gait modalities into one multi-modal representation. In cases where surveillance

Table 15 Comparison with existing person re-identification methods on MARS.

Method	Rank-1 (%)
TLST[84]	61.66
SATPP[85]	69.69
DSAN[86]	69.70
SFT[87]	70.00
IDE(R)+(RK)[88]	70.51
Proposed Multi-Modal Fusion	**73.17**



(A) Sample query set **(B)** Additional query set

Fig. 11 Qualitative retrieval outputs of the multi-modal pipeline, shown as anonymized schematic renderings rather than the original photographs to protect subject identity. Panel (**A**) shows a representative sample query set and panel (**B**) shows an additional query set under different pose, viewpoint, and illumination conditions. In each panel, the left-hand pictogram is the query and the following columns show the top-5 retrieved gallery pictograms; garment colors are sampled from the corresponding original images and pose skeletons are schematic. Green and red boxes indicate correct and incorrect matches, respectively.

video is used, information from the gait modality acts as another temporal cue, while FaceNet embeddings based on MTCNN aligned faces provide identification cues as well when facial parts are present in the image.

Moreover, the results of the experiment show that the proposed model ensemble framework generalises well not only in terms of image-based but also video-based person re-identification benchmarks. As opposed to the current state-of-the-art approaches which mostly rely on the use of appearance-related features, our approach allows for complementing facial embeddings with information related to gait and overall appearance. This allows for increased robustness of the proposed algorithm in terms of practical surveillance scenarios involving, for example, motion blur, partial occlusions, low resolution of pedestrian crops, and similar clothing worn by different pedestrians. In particular, the use of gait information compensates for the lack of information about face region, while MTCNN-FaceNet stream adds more biometric data for close-view pedestrian crops.

Qualitative retrieval analysis

To provide visual evidence of the applicability of the proposed Ensemble Concatenation approach, qualitative retrieval examples are illustrated in Fig. 11, with the representative sample query set shown in Fig. 11a and the additional query set shown in Fig. 11b. To protect subject identity, individuals are rendered as anonymized

schematic pictograms rather than the original photographs: garment color is preserved from the source imagery, since it is a primary discriminative cue used by the model, while facial and other identifying detail is removed and pose is represented schematically. In each panel, the extreme-left pictogram is the query, followed by the top-5 ranked gallery pictograms; green borders indicate correct matches, while red borders indicate incorrect matches.

As shown in Fig. 11a, the multi-modal system retrieves the correct identity at the top ranks across visually diverse queries and remains robust to ambiguities such as similar clothing and partial occlusion. Figure 11b further shows that the retrieval output remains stable under viewpoint, pose, and illumination shifts. These examples demonstrate that combining gait and facial features with global appearance produces a discriminative identity signature for unconstrained CCTV scenes.

Discussion

Interpreting the multi-modal gains

Experimental findings from all three tests somehow keep validating the fundamental hypothesis that a specific biometric modality alone may not be sufficient to ensure successful re-identification of persons in an unconstrained environment, since it fails to encompass everything. On the other hand, there are definitely some improvements in performance by applying proper fusion of these complementary sources of information. For Market-1501, the Ensemble Concatenation model boosts Rank-1 accuracy by 4.21 percentage points compared to the best standalone modality OSNet at 79.80%, and that gap gets larger to 6.56 points in the video based MARS setting. There, temporal motion cues end up being more discriminative than you would expect, honestly. The gain on CUHK03 (2.30 points over OSNet) is more modest, suggesting that viewpoint diversity limits the contribution of the gait branch when only a small number of frames per tracklet is available.

The pattern of individual-modality performance is itself revealing. OSNet consistently produces the strongest single-modality results because the benchmark datasets are dominated by appearance variation rather than deliberate "confuser" scenarios. FaceNet (MTCNN-aligned) achieves comparatively lower standalone scores, which reflects the fundamental difficulty of facial biometrics at the resolution and pose diversity typical of Re-ID benchmarks not a deficiency of the MTCNN alignment step. Indeed, the alignment gate ensures that when a usable face is detected, the resulting embedding is geometrically clean; the bottleneck is the prevalence of undetectable faces due to viewpoint or occlusion. GaitSet's pronounced weakness on CUHK03 (41.5% Rank-1) versus Market-1501 (67.4%) arises because the CUHK03 protocol includes very short tracklets and severe cross-view changes, both of which undermine set-pooling-based gait aggregation.

Cross-dataset generalisation analysis

To address the concern that generalisation was unclear under a single-dataset protocol, we performed an additional cross-dataset evaluation in which the MARS test split was evaluated using the Market-1501-trained weights without dataset-specific fine-tuning. This setting is intentionally more challenging than within-dataset evaluation because the gallery and query distributions differ in camera geometry, bounding-box quality, illumination, and tracklet statistics. The resulting performance is reported in Table 16.

Table 16 Cross-Dataset Generalisation Performance Using Market-1501 Weights on MARS.

Cross-Dataset Metric	Value
Generalisation mAP	56.61%
Generalisation Rank-1	63.27%
Generalisation Rank-5	76.76%
Generalisation Rank-10	81.22%

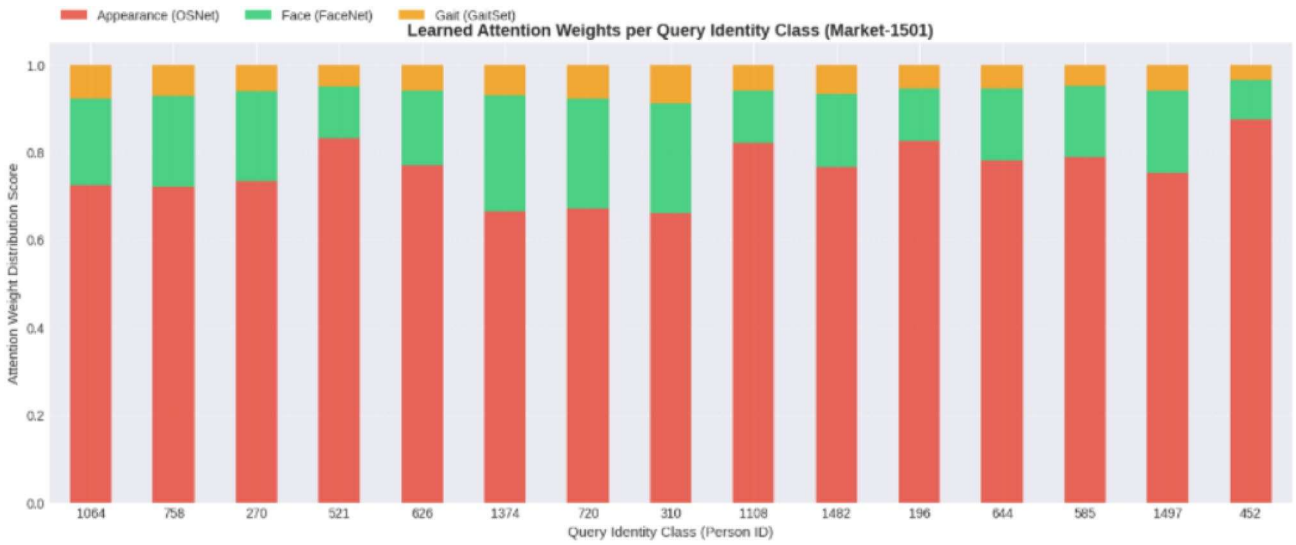


Fig. 12 Ablation of learned attention weights across representative query classes. The grouped bars show how the attention-based ensemble distributes importance across the appearance, face, and gait streams for different query conditions, revealing that the learned fusion often assigns disproportionately high weight to a dominant modality instead of maintaining balanced multi-modal evidence.

The cross-dataset results demonstrate that the proposed framework retains meaningful retrieval capability under domain shift, achieving 56.61% mAP and 63.27% Rank-1 without retraining on the target dataset. The higher Rank-5 and Rank-10 accuracies further indicate that, even when the top-ranked prediction is affected by dataset bias, the correct identity remains within the leading retrieval candidates in most cases. These findings support the generalisation ability of the multi-modal descriptor and show that the fusion of appearance, face, and gait cues transfers beyond the original training distribution.

The role of MTCNN in improving facial embedding quality

A key architectural improvement in this work is the introduction of MTCNN as a explicit detection alignment gate, going just before FaceNet. Prior approaches basically run FaceNet directly on full-body person crops, so you end up with lots of noise coming from background bits, clothing texture, and non facial body parts. That mixing kinda messes with inter class variability in the facial embedding space and it also weakens how complementary the face modality is compared to the appearance modality, because both would end up partially encoding the same visual cues.

By constraining FaceNet input to MTCNN-aligned 160×160 face patches, the facial stream extracts purely biometric information. The confidence-thresholded fallback ($\tau_{\text{face}} = 0.90$) further ensures that only high-quality detections contribute to the fusion vector, trading breadth of coverage for precision of the facial signal. The net effect is that the facial modality provides genuine complementarity to the appearance and gait streams, particularly in "confuser" scenarios where individuals wear identical clothing.

Fusion strategy analysis

Among the six fusion strategies evaluated in Table 12, simple concatenation and weighted averaging deliver the most consistently competitive results. The attention-based ensemble achieves the highest Rank-5 and Rank-10 scores (83.23% and 86.18%, respectively), and also the highest Rank-1 accuracy among all strategies at 74.49%; however, its mAP of 65.10% is substantially below the other ensemble methods. To further investigate this behaviour, Fig. 12 visualises the attention coefficients learned for representative query classes and modality conditions.

The distribution shows that the attention module assigns a disproportionately large weight to the strongest available stream for several query categories while suppressing complementary cues. This supports the interpretation that attention weighting can over-commit to a single dominant modality per query at the expense of holistic retrieval precision: it may find the correct identity quickly when the dominant modality is reliable, but it accumulates more errors across the full gallery. Concatenation avoids this collapse by preserving the contributions of all three modalities without learned suppression, resulting in a more balanced profile across evaluation thresholds. For this reason, concatenation was selected as the main fusion architecture for deployment.

The MLP based fusion gets something like competitive Rank-1 (73.34%) but then the mAP drops to 68.50%. This sort of matches the usual over-fitting tendency we see with learned combiners, especially when the training data is scarce compared with the combined embedding dimensionality. Going forward, it would be good to re-check the MLP based as well as the attention based fusion setups, but now with larger training sets, plus stronger regularisation strategies or whatever works well.

Open-set deployment considerations

Even though the benchmark evaluation in this study is kindof basically following the standard closed-set retrieval protocol, in real-world forensic deployment it usually ends up needing open-set handling where the query identity might not show up in the gallery at all. The proposed framework makes room for this messy situation with a similarity threshold mechanism, which is already laid out in Table 5. Concretely, once the cosine similarity is computed between the query descriptor and each gallery descriptor, the system only accepts the top-ranked candidate if its similarity is above the set threshold ($\tau_{\text{sim}} = 0.30$). If, for some reason, no gallery identity manages to reach that threshold then the query is treated as an unknown identity, instead of being pushed into the nearest gallery match. This modification helps make the retrieval module appear more like a search technique that is verification-aware as opposed to just a closed set ranker and it also ensures that false positive errors where an individual is not in the watchlist do not occur. It also allows the tuning of the threshold depending on how the system will be deployed, where a higher value improves the chances of precision and more conservative forensic alerts, while a lower one focuses on recall.

Limitations and boundary conditions

Several limitations of the current system should be acknowledged.

Even though MTCNN substantially improves facial embedding quality, the system kind of provides no real mechanism for explicitly dealing with partial face occlusions, like masks, glasses, or even that awkward glare. In the future integration phase, an occlusion-aware landmark estimator, or maybe a masked face recognition backbone would help broaden the coverage, especially in healthcare, and security contexts where face covering is pretty common.

Second, even tho the current implementation gets roughly 15–17 FPS with the set inference parameters, some more optimisation would still need to happen if you'aiming for a higher frame rate on multi camera setups. In practice, speedups are often possible by moving the whole thing on the GPU, using mixed precision inference doing model quantisation and pruning too, plus an optimized deployment backend.

Finally, from an ethical standpoint, multi modal biometric surveillance systems carry pretty serious privacy implications, because once you link face, gait, and even appearance embeddings into one long lived identity descriptor, it can really expand re identification reach, more than any one modality by itself. In practice, deployments like these should be governed by the right legal frameworks like proper data minimisation principles, you know the kind that actually matter. And also there should be clear transparency obligations toward the individuals being monitored, not just vague notices that sort of float around.

Conclusion

This paper kind of presents an end to end, multi modal person re identification pipeline that tackles identity ambiguity that is inherent in unconstrained CCTV environments, by doing a kind of principled fusion of three complementary biometric signals. First the omni scale appearance signal, uses OSNet in $\mathbb{R}^{1024}$, then there is MTCNN gated facial geometry with FaceNet in $\mathbb{R}^{128}$, and finally temporal motion patterns, handled by GaitSet in $\mathbb{R}^{256}$. All of that gets unified into a single $\mathbb{R}^{1408}$ Ensemble Concatenation descriptor, which feels like a straightforward arrangement but it really is not. The intro of MTCNN as an explicit face detection and alignment gate is a pretty concrete architectural improvement compared to earlier pipelines that simply plugged face embedding networks onto unaligned full body crops, without any real gating or similar alignment step.

Experiments on Market-1501, MARS, and CUHK03 demonstrated consistent improvements over all three individual modalities. The Ensemble Concatenation model achieved Rank-1/mAP scores of 84.01%/71.46%, 73.17%/69.95%, and 73.50%/69.80% on the three benchmarks respectively, surpassing the best standalone modality in every case. An ablation over six fusion strategies confirmed that concatenation provides the best Rank-1 accuracy and most balanced precision-recall profile for real-world surveillance deployment.

Three directions are identified for future work. First, incorporating a cross-modal Transformer fusion layer learning dynamic attention weights between modality streams conditioned on scene context is expected to improve discriminability beyond what static concatenation can achieve. Second, replacing RGB-based gait estimation with a dedicated silhouette or skeleton extraction module will isolate motion cues from appearance noise and improve gait discriminability particularly on short tracklets. Third, extending the pipeline with an occlusion-aware face alignment module will improve facial-stream coverage in scenarios where partial face covering (masks, glasses) reduces MTCNN detection confidence. Together, these advances are expected to push the system towards the performance levels required for operational forensic deployment at scale.

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Author contributions Swati Shinde (S.S.) led the overall project, including idea generalisation, manuscript preparation, project conceptualisation, research direction, and supervision throughout all stages of the work. Sahil Hirve (S.H.) contributed to idea formulation, implementation of the MTCNN-gated FaceNet facial biometric stream, experimentation, and manuscript preparation. Tushar Badlani (T.B.) was responsible for the design and implementation of the GaitSet-inspired gait recognition module, experimentation, and manuscript preparation. Shashank Dudhane (S.D.) contributed to the development of the person detection and tracking pipeline, as well as manuscript preparation. Saurabh Dhakite (S.D.) implemented the OSNet appearance stream, conducted ensemble fusion experiments, and contributed to manuscript preparation. Lubna A. Gabralla (L.G.) provided guidance throughout the project and contributed to project conceptualisation, manuscript review, figure design, and preparation of the rebuttal and revision documents. Tanupriya Choudhury (T.C.) provided guidance, contributed to result interpretation, conducted the final review, supported code and design updates, formatted the manuscript as per journal guidelines, and updated the rebuttal sheet based on reviewer feedback. Minakshi (M.N.) provided guidance and contributed to experimental analysis, final manuscript review, rebuttal document updates, grammatical corrections, reference and citation management, table design, dataset interpretation, code repository updates on GitHub, and final validation.

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Data availability The datasets analysed during the current study are available from the following public benchmark repositories: Market-1501: https://zheng-lab-anu.github.io/Project/project_reid.html MARS: https://zheng-lab-anu.github.io/Project/project_mars.html CUHK03: https://www.ee.cuhk.edu.hk/~xgwang/CUHK_identification.html

Code availability The source code for the proposed framework is available at https://github.com/tushar-badlani/Project.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval There were no human or animal subjects in the full procedure of the research.

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